

Network_Anomaly_Detection

September 27, 2026

```
[13]: #Importing Libraries
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, \
    f1_score, confusion_matrix, classification_report
import seaborn as sns
import matplotlib.pyplot as plt

[14]: #Defining Column Names and File Path

[15]: # Set the file path to the dataset
file_path = r'KDD+.txt'

# Define the column names corresponding to the NSL-KDD dataset
columns = [
    'duration', 'protocol_type', 'service', 'flag', 'src_bytes', 'dst_bytes',
    'land', 'wrong_fragment', 'urgent', 'hot', 'num_failed_logins', 'logged_in',
    'num_compromised', 'root_shell', 'su_attempted', 'num_root', \
    'num_file_creations',
    'num_shells', 'num_access_files', 'num_outbound_cmds', 'is_host_login', \
    'is_guest_login',
    'count', 'srv_count', 'serror_rate', 'srv_serror_rate', 'rerror_rate', \
    'srv_rerror_rate',
    'same_srv_rate', 'diff_srv_rate', 'srv_diff_host_rate', 'dst_host_count', \
    'dst_host_srv_count',
    'dst_host_same_srv_rate', 'dst_host_diff_srv_rate', \
    'dst_host_same_src_port_rate',
    'dst_host_srv_diff_host_rate', 'dst_host_serror_rate', \
    'dst_host_srv_serror_rate',
    'dst_host_rerror_rate', 'dst_host_srv_rerror_rate', 'attack', 'level'
]

[16]: #Reading the Dataset into a DataFrame
```

```
[17]: # Read the combined NSL-KDD dataset into a DataFrame
df = pd.read_csv(file_path, names=columns)
```

```
[18]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148517 entries, 0 to 148516
Data columns (total 43 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   duration                             148517 non-null  int64
1   protocol_type                        148517 non-null  object
2   service                             148517 non-null  object
3   flag                                 148517 non-null  object
4   src_bytes                           148517 non-null  int64
5   dst_bytes                           148517 non-null  int64
6   land                                148517 non-null  int64
7   wrong_fragment                      148517 non-null  int64
8   urgent                             148517 non-null  int64
9   hot                                 148517 non-null  int64
10  num_failed_logins                   148517 non-null  int64
11  logged_in                           148517 non-null  int64
12  num_compromised                     148517 non-null  int64
13  root_shell                          148517 non-null  int64
14  su_attempted                       148517 non-null  int64
15  num_root                            148517 non-null  int64
16  num_file_creations                  148517 non-null  int64
17  num_shells                          148517 non-null  int64
18  num_access_files                    148517 non-null  int64
19  num_outbound_cmds                   148517 non-null  int64
20  is_host_login                       148517 non-null  int64
21  is_guest_login                      148517 non-null  int64
22  count                              148517 non-null  int64
23  srv_count                           148517 non-null  int64
24  serror_rate                         148517 non-null  float64
25  srv_serror_rate                     148517 non-null  float64
26  rerror_rate                         148517 non-null  float64
27  srv_rerror_rate                     148517 non-null  float64
28  same_srv_rate                       148517 non-null  float64
29  diff_srv_rate                       148517 non-null  float64
30  srv_diff_host_rate                  148517 non-null  float64
31  dst_host_count                      148517 non-null  int64
32  dst_host_srv_count                  148517 non-null  int64
33  dst_host_same_srv_rate              148517 non-null  float64
34  dst_host_diff_srv_rate              148517 non-null  float64
35  dst_host_same_src_port_rate         148517 non-null  float64
36  dst_host_srv_diff_host_rate         148517 non-null  float64
37  dst_host_serror_rate                148517 non-null  float64
```

```

38 dst_host_srv_serror_rate      148517 non-null float64
39 dst_host_rerror_rate          148517 non-null float64
40 dst_host_srv_rerror_rate      148517 non-null float64
41 attack                        148517 non-null object
42 level                        148517 non-null int64
dtypes: float64(15), int64(24), object(4)
memory usage: 48.7+ MB

```

```
[19]: df.tail()
```

```

[19]:      duration protocol_type  service flag  src_bytes  dst_bytes  land  \
148512         0          tcp      smtp  SF         794         333      0
148513         0          tcp      http  SF         317         938      0
148514         0          tcp      http  SF        54540        8314      0
148515         0          udp  domain_u  SF          42          42      0
148516         0          tcp    sunrpc  REJ           0           0      0

      wrong_fragment  urgent  hot  ...  dst_host_same_srv_rate  \
148512              0       0   0  ...              0.72
148513              0       0   0  ...              1.00
148514              0       0   2  ...              1.00
148515              0       0   0  ...              0.99
148516              0       0   0  ...              0.08

      dst_host_diff_srv_rate  dst_host_same_src_port_rate  \
148512              0.06              0.01
148513              0.00              0.01
148514              0.00              0.00
148515              0.01              0.00
148516              0.03              0.00

      dst_host_srv_diff_host_rate  dst_host_serror_rate  \
148512              0.01              0.01
148513              0.01              0.01
148514              0.00              0.00
148515              0.00              0.00
148516              0.00              0.00

      dst_host_srv_serror_rate  dst_host_rerror_rate  \
148512              0.0              0.00
148513              0.0              0.00
148514              0.0              0.07
148515              0.0              0.00
148516              0.0              0.44

      dst_host_srv_rerror_rate  attack  level
148512              0.00  normal      21

```

148513	0.00	normal	21
148514	0.07	back	15
148515	0.00	normal	21
148516	1.00	mscan	14

[5 rows x 43 columns]

```
[20]: # Binary classification target
# Maps normal traffic to 0 and any type of attack to 1
df['attack_flag'] = df['attack'].apply(lambda a: 0 if a == 'normal' else 1)
```

```
[21]: df.tail()
```

```
[21]:
```

	duration	protocol_type	service	flag	src_bytes	dst_bytes	land	\
148512	0	tcp	smtp	SF	794	333	0	
148513	0	tcp	http	SF	317	938	0	
148514	0	tcp	http	SF	54540	8314	0	
148515	0	udp	domain_u	SF	42	42	0	
148516	0	tcp	sunrpc	REJ	0	0	0	

	wrong_fragment	urgent	hot	...	dst_host_diff_srv_rate	\
148512	0	0	0	...	0.06	
148513	0	0	0	...	0.00	
148514	0	0	2	...	0.00	
148515	0	0	0	...	0.01	
148516	0	0	0	...	0.03	

	dst_host_same_src_port_rate	dst_host_srv_diff_host_rate	\
148512	0.01	0.01	
148513	0.01	0.01	
148514	0.00	0.00	
148515	0.00	0.00	
148516	0.00	0.00	

	dst_host_serror_rate	dst_host_srv_serror_rate	dst_host_rerror_rate	\
148512	0.01	0.0	0.00	
148513	0.01	0.0	0.00	
148514	0.00	0.0	0.07	
148515	0.00	0.0	0.00	
148516	0.00	0.0	0.44	

	dst_host_srv_rerror_rate	attack	level	attack_flag
148512	0.00	normal	21	0
148513	0.00	normal	21	0
148514	0.07	back	15	1
148515	0.00	normal	21	0
148516	1.00	mscan	14	1

[5 rows x 44 columns]

```
[23]: print(df[df.attack == 'neptune'])
```

	duration	protocol_type	service	flag	src_bytes	dst_bytes	land	\
2	0	tcp	private	S0	0	0	0	
5	0	tcp	private	REJ	0	0	0	
6	0	tcp	private	S0	0	0	0	
7	0	tcp	private	S0	0	0	0	
8	0	tcp	remote_job	S0	0	0	0	
...	...	...	...	...	...	...		
148485	0	tcp	auth	S0	0	0	0	
148503	0	tcp	private	S0	0	0	0	
148504	0	tcp	private	S0	0	0	0	
148507	0	tcp	private	REJ	0	0	0	
148509	0	tcp	iso_tsap	REJ	0	0	0	

	wrong_fragment	urgent	hot	...	dst_host_diff_srv_rate	\
2	0	0	0	...	0.05	
5	0	0	0	...	0.07	
6	0	0	0	...	0.05	
7	0	0	0	...	0.07	
8	0	0	0	...	0.05	
...	...	...	...	...	...	
148485	0	0	0	...	0.08	
148503	0	0	0	...	0.08	
148504	0	0	0	...	0.07	
148507	0	0	0	...	0.07	
148509	0	0	0	...	0.05	

	dst_host_same_src_port_rate	dst_host_srv_diff_host_rate	\
2	0.0	0.0	
5	0.0	0.0	
6	0.0	0.0	
7	0.0	0.0	
8	0.0	0.0	
...	...	...	
148485	0.0	0.0	
148503	0.0	0.0	
148504	0.0	0.0	
148507	0.0	0.0	
148509	0.0	0.0	

	dst_host_serror_rate	dst_host_srv_serror_rate	dst_host_rerror_rate	\
2	1.00	1.00	0.00	
5	0.00	0.00	1.00	
6	1.00	1.00	0.00	

7	1.00	1.00	0.00
8	1.00	1.00	0.00
...	...	...	...
148485	1.00	1.00	0.00
148503	0.79	0.29	0.21
148504	1.00	1.00	0.00
148507	0.00	0.00	1.00
148509	0.00	0.00	1.00

	dst_host_srv_rerror_rate	attack	level	attack_flag
2	0.00	neptune	19	1
5	1.00	neptune	21	1
6	0.00	neptune	21	1
7	0.00	neptune	21	1
8	0.00	neptune	21	1
...	...	...	...	...
148485	0.00	neptune	18	1
148503	0.71	neptune	21	1
148504	0.00	neptune	21	1
148507	1.00	neptune	21	1
148509	1.00	neptune	20	1

[45871 rows x 44 columns]

```
[24]: # Multi-class classification target categories
dos_attacks = ['apache2', 'back', 'land', 'neptune', 'mailbomb', 'pod',
               'processtable', 'smurf', 'teardrop', 'udpstorm', 'worm']
probe_attacks = ['ipsweep', 'mscan', 'nmap', 'portsweep', 'saint', 'satan']
privilege_attacks = ['buffer_overflow', 'loadmdoule', 'perl', 'ps',
                     'rootkit', 'sqlattack', 'xterm']
access_attacks = ['ftp_write', 'guess_passwd', 'http_tunnel', 'imap',
                  'multihop', 'named', 'phf', 'sendmail', 'snmpgetattack',
                  'snmpguess', 'spy', 'warezclient', 'warezmaster',
                  'xclock', 'xsnoop']

def map_attack(attack):
    if attack in dos_attacks:
        return 1
    elif attack in probe_attacks:
        return 2
    elif attack in privilege_attacks:
        return 3
    elif attack in access_attacks:
        return 4
    else:
        return 0
```

```
# Assign multi-class category to each row
df['attack_map'] = df['attack'].apply(map_attack)
```

```
[27]: df.head(20)
```

```
[27]:
```

	duration	protocol_type	service	flag	src_bytes	dst_bytes	land	\
0	0	tcp	ftp_data	SF	491	0	0	
1	0	udp	other	SF	146	0	0	
2	0	tcp	private	S0	0	0	0	
3	0	tcp	http	SF	232	8153	0	
4	0	tcp	http	SF	199	420	0	
5	0	tcp	private	REJ	0	0	0	
6	0	tcp	private	S0	0	0	0	
7	0	tcp	private	S0	0	0	0	
8	0	tcp	remote_job	S0	0	0	0	
9	0	tcp	private	S0	0	0	0	
10	0	tcp	private	REJ	0	0	0	
11	0	tcp	private	S0	0	0	0	
12	0	tcp	http	SF	287	2251	0	
13	0	tcp	ftp_data	SF	334	0	0	
14	0	tcp	name	S0	0	0	0	
15	0	tcp	netbios_ns	S0	0	0	0	
16	0	tcp	http	SF	300	13788	0	
17	0	icmp	eco_i	SF	18	0	0	
18	0	tcp	http	SF	233	616	0	
19	0	tcp	http	SF	343	1178	0	

	wrong_fragment	urgent	hot	...	dst_host_same_src_port_rate	\
0	0	0	0	...	0.17	
1	0	0	0	...	0.88	
2	0	0	0	...	0.00	
3	0	0	0	...	0.03	
4	0	0	0	...	0.00	
5	0	0	0	...	0.00	
6	0	0	0	...	0.00	
7	0	0	0	...	0.00	
8	0	0	0	...	0.00	
9	0	0	0	...	0.00	
10	0	0	0	...	0.00	
11	0	0	0	...	0.00	
12	0	0	0	...	0.12	
13	0	0	0	...	1.00	
14	0	0	0	...	0.00	
15	0	0	0	...	0.00	
16	0	0	0	...	0.01	
17	0	0	0	...	1.00	
18	0	0	0	...	0.02	

19	0	0	0	...	0.01
----	---	---	---	-----	------

	dst_host_srv_diff_host_rate	dst_host_serror_rate \
0	0.00	0.00
1	0.00	0.00
2	0.00	1.00
3	0.04	0.03
4	0.00	0.00
5	0.00	0.00
6	0.00	1.00
7	0.00	1.00
8	0.00	1.00
9	0.00	1.00
10	0.00	0.00
11	0.00	1.00
12	0.03	0.00
13	0.20	0.00
14	0.00	1.00
15	0.00	1.00
16	0.02	0.00
17	1.00	0.00
18	0.03	0.00
19	0.04	0.00

	dst_host_srv_serror_rate	dst_host_rerror_rate	dst_host_srv_rerror_rate \
0	0.00	0.05	0.00
1	0.00	0.00	0.00
2	1.00	0.00	0.00
3	0.01	0.00	0.01
4	0.00	0.00	0.00
5	0.00	1.00	1.00
6	1.00	0.00	0.00
7	1.00	0.00	0.00
8	1.00	0.00	0.00
9	1.00	0.00	0.00
10	0.00	1.00	1.00
11	1.00	0.00	0.00
12	0.00	0.00	0.00
13	0.00	0.00	0.00
14	1.00	0.00	0.00
15	1.00	0.00	0.00
16	0.00	0.00	0.00
17	0.00	0.00	0.00
18	0.00	0.02	0.00
19	0.00	0.00	0.00

attack	level	attack_flag	attack_map
--------	-------	-------------	------------

0	normal	20	0	0
1	normal	15	0	0
2	neptune	19	1	1
3	normal	21	0	0
4	normal	21	0	0
5	neptune	21	1	1
6	neptune	21	1	1
7	neptune	21	1	1
8	neptune	21	1	1
9	neptune	21	1	1
10	neptune	21	1	1
11	neptune	21	1	1
12	normal	21	0	0
13	warezclient	15	1	4
14	neptune	19	1	1
15	neptune	18	1	1
16	normal	21	0	0
17	ipsweep	18	1	2
18	normal	21	0	0
19	normal	21	0	0

[20 rows x 45 columns]

```
[38]: # Encoding categorical variables
features_to_encode = ['protocol_type', 'service']
encoded = pd.get_dummies(df[features_to_encode])
#df is unchanged here, it is stored in encoded
```

```
[31]: df.head()
```

```
[31]:
```

	duration	protocol_type	service	flag	src_bytes	dst_bytes	land	\
0	0	tcp	ftp_data	SF	491	0	0	
1	0	udp	other	SF	146	0	0	
2	0	tcp	private	S0	0	0	0	
3	0	tcp	http	SF	232	8153	0	
4	0	tcp	http	SF	199	420	0	

	wrong_fragment	urgent	hot	...	dst_host_same_src_port_rate	\
0	0	0	0	...	0.17	
1	0	0	0	...	0.88	
2	0	0	0	...	0.00	
3	0	0	0	...	0.03	
4	0	0	0	...	0.00	

	dst_host_srv_diff_host_rate	dst_host_serror_rate	\
0	0.00	0.00	
1	0.00	0.00	

2	0.00	1.00
3	0.04	0.03
4	0.00	0.00

	dst_host_srv_error_rate	dst_host_error_rate	dst_host_srv_error_rate \
0	0.00	0.05	0.00
1	0.00	0.00	0.00
2	1.00	0.00	0.00
3	0.01	0.00	0.01
4	0.00	0.00	0.00

	attack	level	attack_flag	attack_map
0	normal	20	0	0
1	normal	15	0	0
2	neptune	19	1	1
3	normal	21	0	0
4	normal	21	0	0

[5 rows x 45 columns]

```
[33]: # Numeric features that capture various statistical properties of the traffic
numeric_features = [
    'duration', 'src_bytes', 'dst_bytes', 'wrong_fragment', 'urgent', 'hot',
    'num_failed_logins', 'num_compromised', 'root_shell', 'su_attempted',
    'num_root', 'num_file_creations', 'num_shells', 'num_access_files',
    'num_outbound_cmds', 'count', 'srv_count', 'error_rate',
    'srv_error_rate', 'rerror_rate', 'srv_rerror_rate', 'same_srv_rate',
    'diff_srv_rate', 'srv_diff_host_rate', 'dst_host_count',
    ↪ 'dst_host_srv_count',
    'dst_host_same_srv_rate', 'dst_host_diff_srv_rate',
    'dst_host_same_src_port_rate', 'dst_host_srv_diff_host_rate',
    'dst_host_error_rate', 'dst_host_srv_error_rate', 'dst_host_rerror_rate',
    'dst_host_srv_rerror_rate'
]
```

```
[34]: # Combine encoded categorical variables and numeric features
train_set = encoded.join(df[numeric_features])

# Multi-class target variable
multi_y = df['attack_map']
```

```
[35]: train_X, test_X, train_y, test_y = train_test_split(train_set, multi_y,
    ↪ test_size=0.2, random_state=1337)
```

```
[39]: print(train_X, train_y)
```

	protocol_type_icmp	protocol_type_tcp	protocol_type_udp	service_IRC \
46323	False	True	False	False

126905	False	False	True	False
5989	False	True	False	False
79922	False	True	False	False
121463	False	True	False	False
...	...	...	...	...
91220	False	True	False	False
71538	False	True	False	False
117415	False	True	False	False
9448	False	True	False	False
3223	False	True	False	False

	service_X11	service_Z39_50	service_aol	service_auth	service_bgp	\
46323	False	False	False	False	False	
126905	False	False	False	False	False	
5989	False	False	False	False	False	
79922	False	False	False	False	False	
121463	False	False	False	False	False	
...	...	...	...	...	...	
91220	False	False	False	False	False	
71538	False	False	False	False	False	
117415	False	False	False	False	False	
9448	False	False	False	False	False	
3223	False	False	False	False	False	

	service_courier	...	dst_host_count	dst_host_srv_count	\
46323	False	...	20	173	
126905	False	...	255	250	
5989	False	...	255	62	
79922	False	...	255	4	
121463	False	...	255	19	
...	...	...	...	...	
91220	False	...	50	255	
71538	False	...	255	255	
117415	False	...	255	255	
9448	False	...	201	2	
3223	False	...	170	255	

	dst_host_same_srv_rate	dst_host_diff_srv_rate	\
46323	1.00	0.00	
126905	0.98	0.01	
5989	0.24	0.02	
79922	0.02	0.07	
121463	0.07	0.07	
...	...	...	
91220	1.00	0.00	
71538	1.00	0.00	
117415	1.00	0.00	
9448	0.01	0.07	

3223	1.00	0.00
------	------	------

	dst_host_same_src_port_rate	dst_host_srv_diff_host_rate \
46323	0.05	0.22
126905	0.01	0.00
5989	0.15	0.00
79922	0.00	0.00
121463	0.00	0.00
...	...	...
91220	0.02	0.01
71538	0.00	0.00
117415	0.00	0.00
9448	0.00	0.00
3223	0.01	0.01

	dst_host_serror_rate	dst_host_srv_serror_rate	dst_host_rerror_rate \
46323	0.00	0.00	0.80
126905	0.00	0.00	0.00
5989	0.28	0.32	0.00
79922	1.00	1.00	0.00
121463	1.00	1.00	0.00
...	...	...	...
91220	0.00	0.00	0.00
71538	0.00	0.00	0.00
117415	0.00	0.00	0.00
9448	1.00	1.00	0.00
3223	0.01	0.00	0.01

	dst_host_srv_rerror_rate
46323	0.98
126905	0.00
5989	0.00
79922	0.00
121463	0.00
...	...
91220	0.00
71538	0.00
117415	0.00
9448	0.00
3223	0.00

[118813 rows x 107 columns]	46323	0
126905	0	
5989	0	
79922	1	
121463	1	
..		
91220	0	

```

71538      0
117415      0
9448       1
3223       0
Name: attack_map, Length: 118813, dtype: int64

```

```

[40]: # Further split the training set into separate training and validation sets
multi_train_X, multi_val_X, multi_train_y, multi_val_y = \
    train_test_split(train_X, train_y, test_size=0.3, random_state=1337)

```

```

[41]: #Training the Model

```

```

[42]: # Train RandomForest model for multi-class classification
rf_model_multi = RandomForestClassifier(random_state=1337)
rf_model_multi.fit(multi_train_X, multi_train_y)

```

```

[42]: RandomForestClassifier(random_state=1337)

```

```

[43]: # Predict and evaluate the model on the validation set
multi_predictions = rf_model_multi.predict(multi_val_X)
accuracy = accuracy_score(multi_val_y, multi_predictions)
precision = precision_score(multi_val_y, multi_predictions, average='weighted')
recall = recall_score(multi_val_y, multi_predictions, average='weighted')
f1 = f1_score(multi_val_y, multi_predictions, average='weighted')
print(f"Validation Set Evaluation:")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print(f"F1-Score: {f1:.4f}")

# Confusion Matrix for Validation Set
conf_matrix = confusion_matrix(multi_val_y, multi_predictions)
class_labels = ['Normal', 'DoS', 'Probe', 'Privilege', 'Access']
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues',
            xticklabels=class_labels,
            yticklabels=class_labels)
plt.title('Network Anomaly Detection - Validation Set')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()

# Classification Report for Validation Set
print("Classification Report for Validation Set:")
print(classification_report(multi_val_y, multi_predictions, \
    target_names=class_labels))

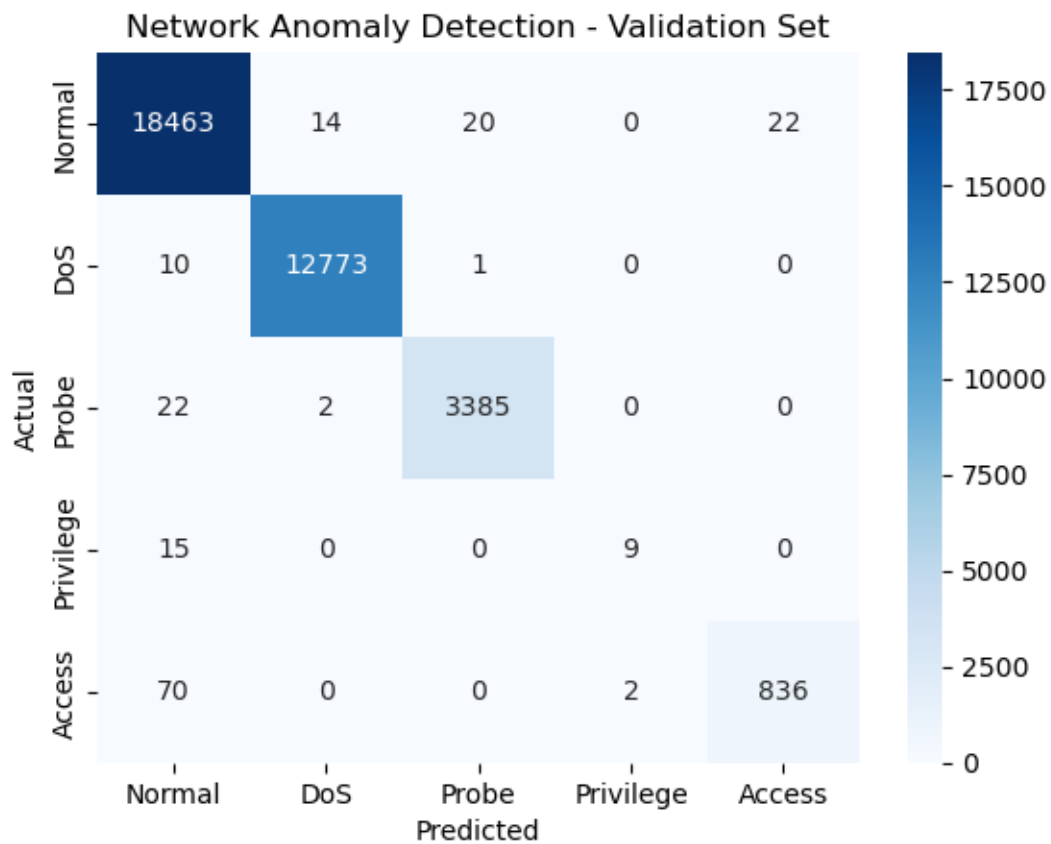
```

```

Validation Set Evaluation:
Accuracy: 0.9950

```

Precision: 0.9949
 Recall: 0.9950
 F1-Score: 0.9949



Classification Report for Validation Set:

	precision	recall	f1-score	support
Normal	0.99	1.00	1.00	18519
DoS	1.00	1.00	1.00	12784
Probe	0.99	0.99	0.99	3409
Privilege	0.82	0.38	0.51	24
Access	0.97	0.92	0.95	908
accuracy			1.00	35644
macro avg	0.96	0.86	0.89	35644
weighted avg	0.99	1.00	0.99	35644

[44]: *#Testing the model on Test set*

```
[45]: test_multi_predictions = rf_model_multi.predict(test_X)
test_accuracy = accuracy_score(test_y, test_multi_predictions)
test_precision = precision_score(test_y, test_multi_predictions,
    ↪average='weighted')
test_recall = recall_score(test_y, test_multi_predictions, average='weighted')
test_f1 = f1_score(test_y, test_multi_predictions, average='weighted')
print("\nTest Set Evaluation:")
print(f"Accuracy: {test_accuracy:.4f}")
print(f"Precision: {test_precision:.4f}")
print(f"Recall: {test_recall:.4f}")
print(f"F1-Score: {test_f1:.4f}")

# Confusion Matrix for Test Set
test_conf_matrix = confusion_matrix(test_y, test_multi_predictions)
sns.heatmap(test_conf_matrix, annot=True, fmt='d', cmap='Blues',
    xticklabels=class_labels,
    yticklabels=class_labels)
plt.title('Network Anomaly Detection')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()

# Classification Report for Test Set
print("Classification Report for Test Set:")
print(classification_report(test_y, test_multi_predictions,
    ↪target_names=class_labels))
```

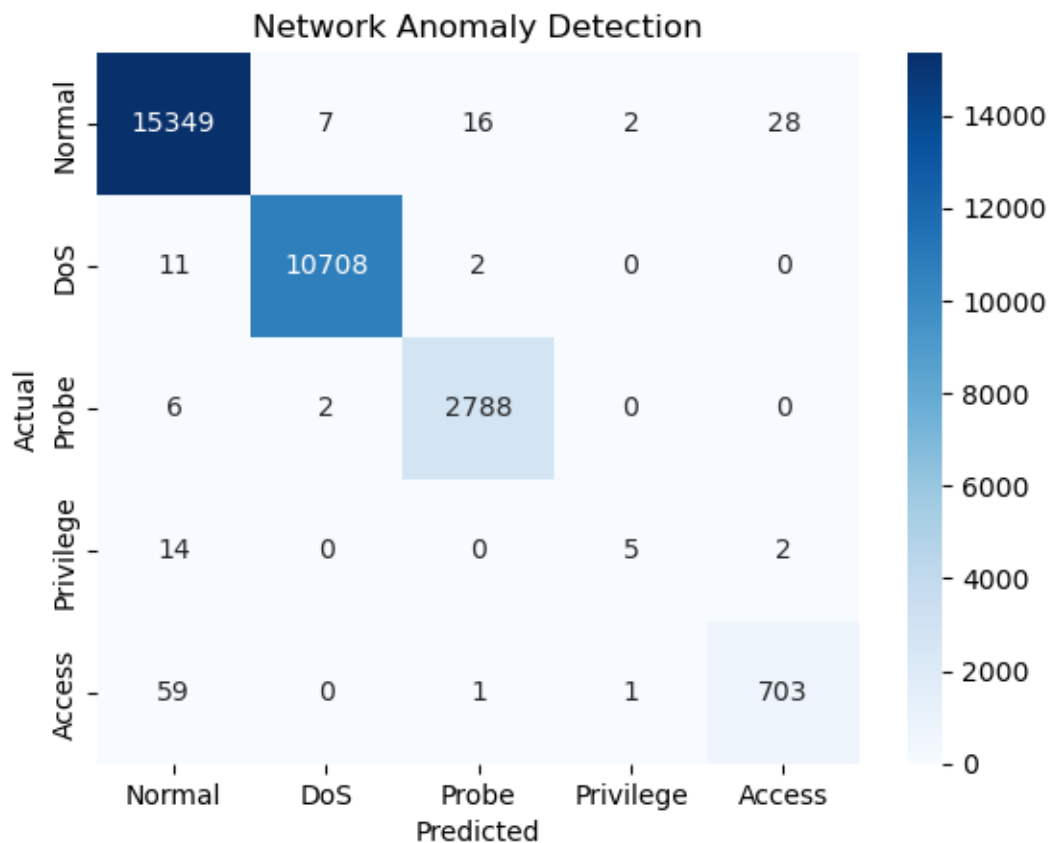
Test Set Evaluation:

Accuracy: 0.9949

Precision: 0.9947

Recall: 0.9949

F1-Score: 0.9947



Classification Report for Test Set:

	precision	recall	f1-score	support
Normal	0.99	1.00	1.00	15402
DoS	1.00	1.00	1.00	10721
Probe	0.99	1.00	1.00	2796
Privilege	0.62	0.24	0.34	21
Access	0.96	0.92	0.94	764
accuracy			0.99	29704
macro avg	0.91	0.83	0.85	29704
weighted avg	0.99	0.99	0.99	29704

```
[46]: import joblib

# Save the trained model to a file
model_filename = 'network_anomaly_detection_model.joblib'
joblib.dump(rf_model_multi, model_filename)
```

```
print(f"Model saved to {model_filename}")
```

Model saved to network_anomaly_detection_model.joblib

[]: